CS6500 — Network Security / security cluster

Tier	B
Phase (order)	6
Do this first	CS3205, CS1202
Estimated effort	10-12 weeks
Unlocks	CS6530, CS6570

Roadmap

1. Cryptography basics, TLS, authentication, firewalls and attacks.
2. Secure systems engineering (CS6570) and applied crypto (CS6530) as follow-ups.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Computer - Cryptography and Network Security (41 videos). NPTEL *Cryptography and Network Security* (Debdeep Mukhopadhyay, IIT Kharagpur).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Introduction to Cryptography Lectures (27 videos). Christof Paar *Introduction to Cryptography* (official channel): the clearest crypto lecturer; uses his free-chapter textbook.
- MIT 6.858 Computer Systems Security, Fall 2014 (22 videos). MIT 6.858 *Computer Systems Security* (Fall 2014, official OCW): buffer overflows, web security, sandboxing.
- Information Security 5 Secure Systems Engineering | IIT Madras (59 videos). NPTEL *Secure Systems Engineering* (Chester Rebeiro, IIT Madras): IITM's own systems security course.

⚡ Short videos for intuition and revision

- Cryptography & Network Security (83 videos). Neso Academy *Cryptography & Network Security*: short revision videos.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–9	Overview, number theory, probability, classical ciphers, Shannon's theory. Do Cryptopals set 1.

Week	Lectures	What to do
2	10–18	Block ciphers, DES, AES, linear and differential cryptanalysis, modes of operation. Break ECB in Cryptopals set 2.
3	19–26	Stream ciphers, pseudorandomness, hash functions, MACs.
4	27–33	RSA, primality, factoring, discrete log, Diffie-Hellman, ElGamal. Implement textbook RSA and a small-exponent attack.
5	34–37	Elliptic curves, secret sharing.
6	38–41	Network protocols, system security, firewalls, IDS, side channels. Then web/TLS security from MIT 6.858 (alt); set up a firewall with nftables.

▼ **All lectures in *Computer - Cryptography and Network Security* (41 videos)**

1. Introduction
2. Overview on Modern Cryptography
3. Introduction to Number Theory
4. Probability and Information Theory
5. Classical Cryptosystems
6. Cryptanalysis of Classical Ciphers
7. Shannons Theory
8. Shannons Theory (Contd...1)
9. Shannons Theory (Contd...2)
10. Symmetric Key Ciphers
11. Block Cipher Standards (DES)
12. Block Cipher Standards (AES)
13. Block Cipher Standards (AES) (Contd.)
14. Linear Cryptanalysis
15. Differential Cryptanalysis
16. Few other Cryptanalytic Techniques
17. Overview on S-Box Design Principles
18. Modes of Operation of Block Ciphers
19. Stream Ciphers
20. Stream Ciphers (Contd...1)
21. Stream Ciphers (Contd...2)
22. Pseudorandomness
23. Cryptographic Hash Functions
24. Cryptographic Hash Functions (Contd...1)
25. Cryptographic Hash Functions (Contd...2)
26. Message Authentication Codes
27. More Number Theoretic Results
28. The RSA Cryptosystem

29. Primality Testing
30. Factoring Algorithms
31. Some Comments on the Security of RSA
32. Discrete Logarithm Problem (DLP)
33. The Diffie-Hellman Problem and Security of ElGamal Systems
34. An Introduction to Elliptic Curve Cryptography
35. Application of Elliptic Curves to Cryptography
36. Implementation of Elliptic Curve Cryptography
37. Secret Sharing Schemes
38. A Tutorial on Network Protocols
39. System Security
40. Firewalls and Intrusion Detection Systems
41. Side Channel Analysis of Cryptographic Implementations

Books, notes and websites

- Book: Boneh & Shoup, *A Graduate Course in Applied Cryptography* (free)
- Stanford Cryptography I (Dan Boneh)
- Book: Katz & Lindell, *Introduction to Modern Cryptography*
- cryptopals.com, picoCTF, OverTheWire

Practice / proof of understanding

Complete cryptopals sets 1-2 and 10 picoCTF challenges.

What your IITM course sheet says

From the PDF: CS6500

Official description: Objectives: The objective of this course is to teach the concepts of securing computer network protocols, based on the application of cryptography techniques.

Official syllabus (trimmed):

1. Basics of cryptography, including conventional and public-key cryptography, hash functions, authentication, and digital signatures. (4 hrs)
2. Key Management and Distribution: Symmetric Key Distribution, Distribution of Public Keys, X.509 Certificates, Public-Key Infrastructure. (2 hrs)
3. User Authentication: Remote User-Authentication Principles, Remote User-Authentication Using Symmetric Encryption, Kerberos Systems, Remote User Authentication Using Asymmetric Encryption. (2 hrs)
4. Malicious Software: Viruses, Worms, System Corruption, Attack Agents, Information Theft — Keyloggers, Phishing, Spyware Payload Stealthing, Backdoors, Rootkits, Distributed Denial of Service Attacks. (4 hrs)
5. Network Access Control: Network Access Control, Extensible Authentication Protocol, IEEE 802.1X Port-Based Network Access Control. (3 hrs)
6. IP Security: IP Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange (IKE). (5 hrs)
7. Transport-Level Security: Web Security Considerations, Secure Sockets Layer, Transport Layer Security, HTTPS standard, Secure Shell (SSH) application. (4 hrs)
8. Electronic Mail Security: Pretty Good Privacy, S/MIME, DomainKeys Identified Mail. (4 hrs)
9. Wireless Network Security: Mobile Device Security, IEEE 802.11i Wireless LAN Security. (4 hrs)
10. Firewalls and Intrusi

Official textbooks: 1. Cryptography and Network Security: Principles and Practice, 6th Edition, William Stallings, 2014, Pearson, ISBN13: 9780133354690.

Official references: 1. Network Security: Private Communications in a Public World, M. Speciner, R. Perlman, C. Kaufman, Prentice Hall, 2002. 2. Linux iptables Pocket Reference, Gregor N. Purdy, O'Reilly, 2004, ISBN-13: 978-0596005696. 3. Linux Firewalls, by Michael Rash, No Starch Press, October 2007, ISBN: 978-1-59327-141-1. 4. Network Security, Firewalls And VPNs, J. Michael Stewart, Jones & Bartlett Learning,

From the PDF: CS6530

Official description: The objectives of the course include learning the basic principles of applied cryptography and the commonly used cryptographic algorithms for encryption, message authentication and integrity, and digital signatures.

Official syllabus (trimmed):

> Unit 1: NUMBER THEORY BASICS: Modular arithmetic, primes, GCD and Chinese reminder theorems. BLOCK CIPHERS: DES, AES; ECB, CBC, OFB,CFB, CTR and GCM modes, Double and triple encryptions. STREAM CIPHERS: Encryption and decryption with Stream ciphers, Shift-register based stream ciphers, Currently used Stream Ciphers. Unit 2: PUBLIC KEY CRYPTOGRAPHY: RSA, ElGamal, Rabins encryption schemes, Diffe-Hellman Key exchange, practical digital signatures. ELLIPTIC CURVE CRYPTOLOGY: definitions, group properties and basic algorithms for group operations. Unit 3: HASH FUNCTIONS: oneway, collision resistant, preimage resistant HASH functions, Real-world examples. Random Number Generation and Secret Sharing. MESSAGE AUTHENTICATION CODES: MAC from Hash functions, MAC from block ciphers. KEY ESTABLISHMENT PROTOCOLS: Diffie-Hellman Protocol, Man-in-the-middle Attack, certificates, Public Key Infrastructure and PKI based crypto systems.

Official textbooks: 1. Christof Paar, Jan Pelzl, Understanding Cryptography, A text book for students and practitioners, Springer Verlag, India, 2010 { INDIAN EDITION} 2. Alfred Menezes. et al, Handbook of Applied Cryptography, CRC Press, USA. { Entire book available for free download in PDF form from authors homepage} 3. William Stallings, “Cryptography and Network Security”, Pearson, 2014.

Official references: None.

From the PDF: CS6111

Official description: 1. Build a solid mathematical basis to understand foundations of cryptography 2. Formally understand the notions related to security authentication and privacy. 3. Provide a rigorous treatment of the emerging and key subject subarea of CSE - security.

Official textbooks: Oded Goldreich, The Foundations of Cryptography: VOL 1 Cambridge university Press, USA. 2001. Oded Goldreich, The Foundations of Cryptography: VOL 2 Cambridge university Press, USA. 2004. (Indian print for both volumes are available).

Official references: J.Katz, Y.Lindell, Introduction to Modern Cryptography, Chapman Hall, USA 2007. Wen Bo Mao, Modern cryptography - Theory and practice, Prentice Hall, USA, 2003 (Indian edition available)

From the PDF: NUS180

The course sheet has no syllabus text for this course; the plan below is based on the standard content of this subject.